Digital Notes Sharing Website

A Project Report

Submitted in partial fulfilment of

the Requirements for the award of the degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

By

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2025 - 2026

PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

PRN No.: ....   Roll no: ____________________

- Name of student

- Title of the Project

- Name of the Guide

- Teaching experience of the Guide

- Is this your first submission?   Yes ☐   No ☐

Signature of the student   Sign of the Guide

Signature of the Co-Ordinator

Date:………………….

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2025-2026

CERTIFICATE

This is to certify that the project entitled, **Digital notes sharing website** is bonafied of work **Sahil Dewan** is bearing seat no.: Submitted in partial fulfilment of the requirements for the award award of degree of **BACHELOR OF SCIENCE** in **INFORMATION TECHNOLOGY** from university of Mumbai.

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External Examiner

Date : **College seal**

ABSTRACT

In modern education, students often face difficulties in sharing and accessing study notes efficiently. Traditional methods like paper notes, photocopies, or emails are often disorganized, time-consuming, and prone to loss. The Digital Notes Sharing Website aims to solve this problem by providing a centralized online platform where students can upload, share, and access study notes anytime and anywhere. This platform allows students to store notes in a structured manner, search for notes based on subjects or topics, and download them securely. It also encourages collaboration among students, as everyone can contribute and benefit from shared resources. With features like user authentication, admin management, and file uploads, the system ensures security and proper organization of all study material. Overall, this project enhances the learning experience by making study notes easily accessible, reducing dependency on physical copies, and promoting a culture of knowledge sharing among students. It is simple, user-friendly, and can be used by students of any technical skill level.

ACKNOWLEDGEMENT

We feel immense pleasure to introduce “ **Digital Notes Sharing Website**” as my project.

We express our foremost and sincere gratitude to Firstly our teachers, and secondly to our family and friends. I Would like to thank you principal “**DR.Santosh Ojha**” & our Head of Department “**Mrs. Asmita Khochre**”. I would like to give a very special honor and respect to our teachers, “**Prof. Ms.Tejaswi kadam**” who took keen interest in checking the minuted details of the project work and guided us throughout the same. Not only from this project but apart from this subject, we have learnt a lot from them, which we are sure will be useful in different stages in the future of our life. We would also like to express our gratitude to also our beloved parents for their review and many helpful comments and enlightening us and guiding us throughout the finalization of this project within the limited time frame. We would also like to underscore dynamic efforts of our teamwork and our contributions towards the preparation of this project. Last but not the least, we thank to all our teachers and as well as friends who have given us that much strength to keep moving on forward every time. We are greatly thankful to one and all and have no words to express our gratitude to them.

DECLARATION

I hereby declare that the project entitled "**Digital Notes Sharing Website**"

Was done at Asmita college, has not in any case been duplicated or submitted to any other university for the award of any degree.

To the best of my knowledge, other than me, no one has submitted this project to any other university. The project is done in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY) to be submitted as final semester project as part of our curriculum.

Mr. Sahil Altab Dewan

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Chapter 1

Introduction

- In today's fast-paced educational environment, students need quick and easy access to study materials. Traditional methods of sharing notes, such as physical copies or emails, are often inefficient, unorganized, and prone to loss. A centralized digital platform can solve these problems by allowing students to upload, share, and access notes online in a structured and secure manner.
- The Digital Notes Sharing Website is designed to provide a simple and user friendly solution for students. It allows users to store their study materials in one place, search for notes based on subjects or topics, and download them whenever needed. The platform also encourages collaboration among students, as everyone can contribute and benefit from shared resources.
- By using modern web technologies like HTML, CSS, JavaScript, PHP, and MySQL, the system ensures that notes are organized, secure, and easily accessible. This project aims to enhance the learning experience, reduce dependency on physical notes, and promote efficient knowledge sharing among students, making study materials available anytime and anywhere.

1.1 Background

- In the modern education system, students often rely on physical notes or personal copies for studying. Sharing these notes with classmates can be difficult, especially if someone misses a lecture or misplaces their notes. Traditional methods like photocopying or emailing files are often inefficient and unorganized.
- With the advancement of digital technology, there is an opportunity to create a centralized platform where students can upload, share, and access study materials online. Such a system helps students find the information they need quickly and keeps notes organized in one place.
- The Digital Notes Sharing Website provides a solution by allowing students to contribute their notes and access notes uploaded by others. This promotes collaboration, saves time, reduces dependency on physical copies, and ensures that study material is always available when required.

1.2 Objectives

- The main objective of the Digital Notes Sharing Website is to provide a centralized and convenient platform for students to upload, share, and access study notes online. Traditional methods of sharing notes, such as physical copies or emails, are often time-consuming and unorganized. By creating this platform, students can store all their study materials in one place, ensuring that important notes are never lost and can be accessed anytime and anywhere.

- **Other specific objectives of the project include:**

1. Enable students to search and retrieve notes based on subjects, topics, or keywords, saving time and effort
2. Promote collaboration and knowledge sharing among students, allowing everyone to benefit from shared resources.
3. Provide a secure environment where only authorized users can upload or download notes, protecting sensitive or personal data.
4. Create a user-friendly interface that is simple to navigate, making it accessible for students with different technical skill levels.
5. Maintain a well-organized database to ensure all notes are stored efficiently and can be managed easily by the system and admin.

Overall, the project aims to enhance the learning experience, reduce dependency on physical notes, and make study materials more accessible, organized, and reliable. By achieving these objectives, students can improve their preparation, collaborate effectively with peers, and have a smoother, more productive academic experience.

1.3 Scope, Applicability and Purpose

Purpose

• To provide a centralized digital platform for students to upload and share study notes easily.

• To reduce dependency on physical notes and avoid loss or damage of important study materials.

• To encourage collaboration and knowledge sharing among students from the same class or course

• To ensure that all notes are organized, secure, and easily retrievable when needed.

• To improve the overall learning experience by providing students with quick access to study materials.

• To make study resources available anytime and anywhere, supporting both classroom and remote learning

Scope

• Students can upload, download, and search notes based on subjects, topics, or keywords.

• Admin can manage users and notes, control access, and maintain system security.

• The system supports multiple file formats, such as PDF, DOCX, and PPT, for flexibility.

• Scalable for future expansion, like adding new subjects, semesters, or departments.

• Useful in colleges, universities, or online learning platforms for centralized note sharing.

Applicability

• Beneficial for students of all academic levels who want easy access to study materials.
• Supports efficient exam preparation and reduces time spent searching for notes.

• Encourages a digital learning culture, reducing paper usage and promoting eco-friendly practices.

• Can be used by educational institutions to improve student collaboration and resource management.

1.4 Achievement

• Development of a fully functional Digital Notes Sharing Website that allows students to upload, download, and share study materials.

• Implementation of a secure login system for both students and admin to protect user data.

• Creation of a centralized database to store notes, user information, and related resources in an organized manner.

• Development of a user-friendly interface that makes navigation simple for students of all technical skill levels.

• Integration of search functionality, allowing students to find notes based on subjects, topics, or keywords quickly.

• Admin module to manage users and notes, ensuring proper control and security of the platform.

• Support for multiple file formats, such as PDF, DOCX, and PPT, providing flexibility for users.

• Promotion of collaboration and knowledge sharing among students, improving learning efficiency.

1.5 Organization of reports

Here's an outline for organizing reports in a Online Shopping Portal :

Report Categories for Digital Notes Sharing Website

User Report:

- Login/logout Account
- Register Account
- Download and preview Notes
- Upload there Note

Notes Report:

- Notes Catalog Overview
- Notes easily download in one click
- Notes by Subject or Topic
- Notes Upload Trends

Admin Report:

- See Total Users on Website
- Notes Approval/Verification Status
- Check User notes
- Upload Notes / delete Notes

Activity Report:

- Daily Notes upload
- Downloadable notes
- Easy preview
- User engagement trends

Report Types

1. Summary Reports of Overview of key metrics (e.g., total users, total notes uploaded, active users).

2. Detail Reports of In-depth analysis of specific data (e.g., individual user activity, notes download history).

3. Comparative Reports o Side-by-side comparisons (e.g., notes uploaded by subject, weekly user activity).

4. Trend Reports o Historical data analysis (e.g., monthly uploads, user growth trends).

Report Frequency

1. Daily Reports of Notes uploaded today o Active users today

2. Weekly Reports of Weekly summary of user activity o Weekly summary of notes uploaded and downloads o Weekly system performance and issues

3. Monthly Reports o Monthly user registration trends o Monthly notes catalog overview of Monthly most active users and most downloaded notes

Chapter 2

Survey Of Technology

2.1 Frontend

The frontend is the part of the Digital Notes Sharing Website that the user interacts with directly. It includes all the visual elements like pages, buttons, forms, and menus.

- **HTML** is used to structure the content on the website, such as headings, paragraphs, forms, and tables.

- **CSS** is used to style the website, including colors, fonts, layouts, and spacing, making it visually appealing and easy to navigate.

- **JavaScript** is used to make the website interactive. It helps in validating forms, handling button clicks, searching notes dynamically, and improving the overall user experience.

Together, **HTML**, **CSS**, and **JavaScript** form the frontend technology stack that ensures the website is user-friendly, responsive, and visually organized.

2.2 Backend

The backend is the part of the Digital Notes Sharing Website that works behind the scenes. It handles all the logic, data processing, and communication between the frontend and the database.

- **PHP** (Hypertext Preprocessor) is used as the backend programming language. It processes user requests, interacts with the database, and generates dynamic content for the frontend.

- **MySQL** is used as the database management system. It stores all data, including user information, uploaded notes, download records, and activity logs, in an organized manner.

- The backend ensures that data is secure, consistent, and accessible only to authorized users.

- It also manages important functionalities like user authentication, notes uploading and downloading, and admin control over users and notes.

- Together, **PHP** and **MySQL** form the backend technology stack, making the website functional, secure, and capable of handling multiple users efficiently.

2.3 Reasons for using This Techniques

• The Digital Notes Sharing Website uses a combination of frontend and backend technologies to ensure the system is user-friendly, secure, and functional. **HTML**, **CSS**, and **JavaScript** are used in the frontend to create a responsive and interactive interface, making it easy for students to navigate, upload, and download notes.

• **PHP** is chosen for the backend because it is widely used, easy to learn, and integrates smoothly with **MySQL** databases. PHP allows the system to process user requests efficiently, handle file uploads, manage sessions, and ensure secure communication between users and the database.

• **MySQL** is used to store all data in a structured way. It is reliable, fast, and capable of handling multiple users simultaneously. Using MySQL ensures that notes, user information, and activity logs are organized and easily retrievable.

• **Visual Studio Code** (VS Code) is used as the development environment. It provides a lightweight, user-friendly editor with features like syntax highlighting, extensions for PHP and MySQL, and integrated terminal support, which makes coding and testing easier without the need for heavy local server setups.

• Overall, this combination of technologies is efficient, cost-effective, and suitable for developing a secure and functional Digital Notes Sharing Website

2.3.1 EXISTING SYSTEM

• In the existing system, students often rely on physical notes, books, or email attachments to share study materials. This method is time-consuming, unorganized, and inefficient.

• Students who miss a class or lose their notes may find it difficult to catch up, as there is no centralized system to access all study materials.

• Sharing notes through email or messaging apps creates duplicate copies, confusion, and difficulty in tracking updates

. • Existing systems do not provide proper search functionality, so students spend more time finding relevant notes based on subjects or topics.

• There is also limited security and control, as anyone with access to emails or shared files can misuse or delete important study materials.

• Overall, the existing system is manual, less collaborative, and does not provide efficient knowledge sharing, which creates a need for a better digital solution.

2.3.2 PROPOSED SYSTEM

• The proposed Digital Notes Sharing Website is designed to provide students with a fast, secure, and convenient way to share and access study materials online. The system aims to overcome the problems of traditional note-sharing methods such as time consumption, lack of organization, and difficulty in accessibility.

• This platform allows students to create accounts, upload their notes, and browse or download notes uploaded by others. It also includes search and filter features, enabling users to quickly find notes related to a specific subject or topic

. • The website will have an admin panel to monitor uploads, manage users, and maintain data integrity.

• The system promotes collaboration, digital learning, and resource sharing among students. It reduces paper use and ensures that study materials are available 24/7 from any device, helping students stay organized and prepared for exams efficiently.

Chapter 3

Requirement and Analysis

3.1 Problem definition (problem in current system)

The current system of sharing notes has several limitations that affect both users (students) and administrators.

1. **User Issues:**

• Difficulty in accessing notes when they are scattered across emails or physical copies.

• Lack of search functionality, making it time-consuming to find relevant notes.

• Inability to track which notes are updated or reliable.

• Notes can be lost or misplaced, especially if shared manually.

• No collaboration or sharing platform, limiting student interaction and knowledge sharing.

2. **Admin Issues**:

• Difficulty in managing users and their activities manually.

• Lack of control over notes quality and authenticity, leading to unverified content.

• Inability to track uploads, downloads, or system usage efficiently.

• Manual handling of issues like duplicate notes or inappropriate uploads.

• Difficulty in maintaining data security and integrity across the system.

3.2 Data Requirement (Data description)

The Digital Notes Sharing Website requires structured data to manage users, notes, and system activities efficiently. The following are the key data requirements:

1. User Data:

- Name, email, and contact details of each student.
- Login credentials including username and password.
- Role information (e.g., admin or student).
- Activity logs like login history, notes uploaded, and downloads.

2. Notes Data:

- Note title, subject, and topic description.
- File type (PDF, DOCX, PPT) and file size.
- Upload date and the user who uploaded the note.
- Download count and access history for each note.

3.Admin Data:

- Admin login credentials and profile information.
- User management data, including status (active/inactive) and permissions.
- System logs for monitoring uploads, downloads, and user activities.

4.System Data:

- Metadata for notes and users to support search and filtering.
- Security data for authentication and authorization.
- Backup and recovery data to prevent loss of information.

This structured data ensures that the system is organized, secure, and able to handle multiple users efficiently.

3.3 Software and Hardware Requirements

The Digital Notes Sharing Website requires both software and hardware to function efficiently. The requirements are divided as follows:

1. Software Requirements:

• Operating System: Windows, Linux, or macOS for development and testing.

• Frontend Technologies: HTML, CSS, and JavaScript for creating responsive and interactive web pages.

• Backend Technology: PHP for processing user requests, handling file uploads, and connecting to the database.

• Database Management System: MySQL for storing user information, notes, and system logs.

• Development Environment: Visual Studio Code (VS Code) for coding, debugging, and testing the application.

• Web Browser: Chrome, Firefox, or Edge for testing and accessing the website.

2. Hardware Requirements:

• Processor: Minimum Intel i3 or equivalent for smooth operation.

• RAM: At least 4 GB for running development tools efficiently.

• Storage: Minimum 500 MB free space for project files and database storage.

• Display: 14-inch or larger screen for better development experience.

• Internet Connection: Required for downloading tools, libraries, and testing online functionalities.

These requirements ensure the system is efficient, reliable, and capable of handling multiple users simultaneously.

3.4 Planning and Scheduling

3.4.1 Planning

• Planning involves identifying all the tasks required to complete the Digital Notes Sharing Website project successfully.

• It includes defining project goals, determining the resources needed, setting timelines, and assigning responsibilities.

• Proper planning ensures that the project progresses systematically and reduces the risk of delays or errors.

• Tasks identified for this project include requirement analysis, system design, database design, frontend and backend development, integration, testing, and deployment.

• Planning also helps in estimating the duration of each task and understanding dependencies between tasks.

3.4.2 Scheduling

• Scheduling involves creating a timeline for completing each task identified in the planning phase.

• It ensures that all activities are completed on time and resources are used efficiently.

• Scheduling is essential for monitoring progress and ensuring that the project stays on track.
• Tools like GANTT Charts are commonly used to represent schedules visually, showing start dates, end dates, and task durations.

• Effective scheduling allows timely completion of tasks, helping the project meet its goals and deadlines.

GANTT Chart :

A Gantt chart is a project management tool assisting in the planning and scheduling of projects of all sizes; they are particularly useful for visualizing projects. A Gantt chart is defined as a graphical representation of activity against time; it helps project professionals monitor progress. They facilitate the creation of complex plans, particularly those involving multiple teams and shifting deadlines. Gantt charts assist teams in scheduling tasks to meet deadlines and allocate resources effectively. This enables you to quickly determine :

• What the various activities are when each activity begins and ends.

• How long each activity is scheduled to last.

• Where activities overlap with other activities, and by how much.

• The start and end date of the whole project.

Creating a Gantt chart for a Digital notes sharing website involves outlining key tasks and their timelines. Here's a simplified version of what that might look like

ID	Activities	Start date	End date	Duration
1	Requirement analysis	01-Oct-2025	03-Oct-2025	3 days
2	System Design	04-Oct-2025	08-Oct-2025	5 days
3	Database Design	09-Oct-2025	11-Oct-2025	3 days
4	Frontend Development	12-Oct-2025	18-Oct-2025	7 days
5	Backend Development	12-Oct-2025	20-Oct-2025	9 days
6	Integration of Frontend & Backend	21-Oct-2025	23-Oct-2025	3 days
7	Testing	24-Oct-2025	28-Oct-2025	5 days
8	Deployment	29-Oct-2025	30-Oct-2025	2 days

3.5 Conceptual Models

Implementation methodology

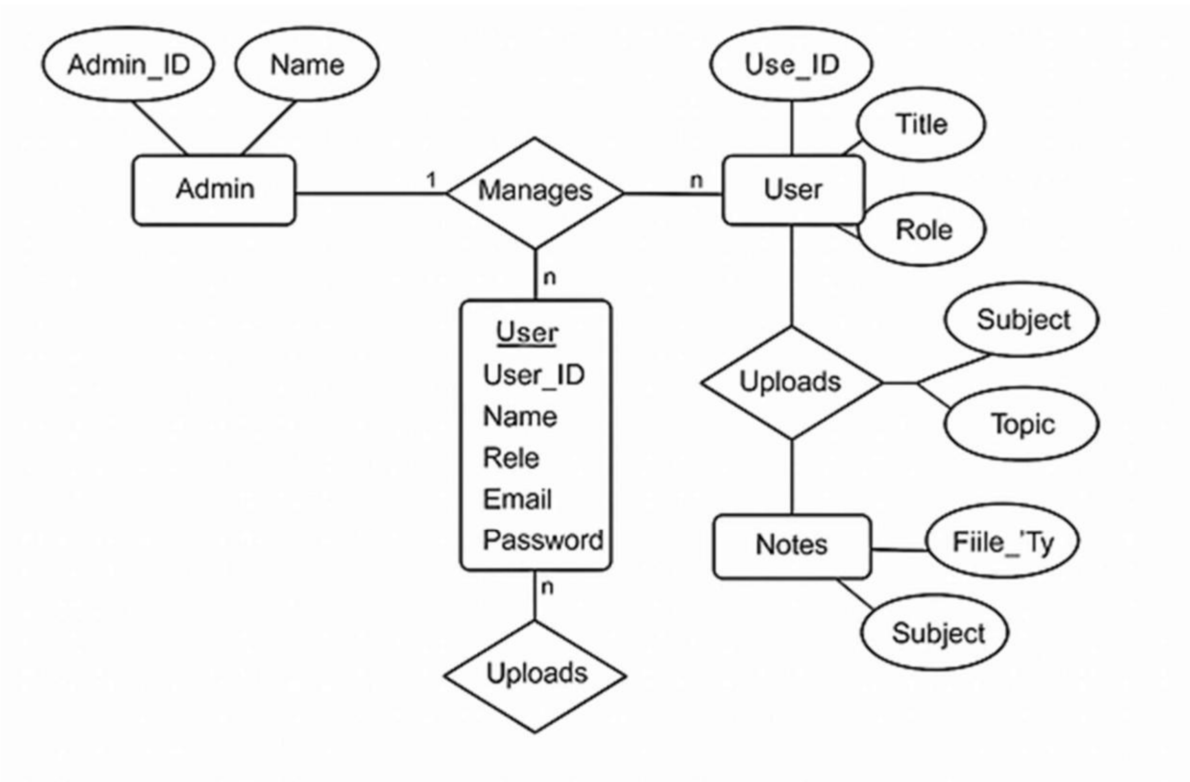
Prototype Model :

The Prototype Model is a software development approach where a preliminary version of the system is built, tested, and refined through multiple iterations. In the context of the Digital Notes Sharing Website, the prototype serves as a visual and functional representation of the platform's core features—such as user registration, note uploading, and downloading. This model allows developers and stakeholders to interact with the system early in the development cycle, providing valuable feedback that guides further improvements. It emphasizes user experience and helps identify potential issues before full-scale development begins.

Purpose of using Prototype Model:

The primary purpose of using the Prototype Model is to reduce misunderstandings between developers and users by offering a tangible preview of the final product. It enables early detection of design flaws, improves requirement clarity, and enhances user involvement. For a student-centric platform like this, prototyping ensures that the interface is intuitive and the features align with academic needs. It also accelerates development by refining requirements through real-time feedback, ultimately leading to a more robust and user-friendly final system.

3.5.1 E-R Diagram (Based on data requirement)



Entity Relationship Diagrams (ERDs) are used to visually represent the structure and layout of a database. They help describe the database schema by identifying key entities (which often correspond to tables) and the relationships between them. ERDs provide a clear conceptual view of the system, independent of hardware or software. While some ERDs map directly to every table in a database, others are more abstract, focusing on major concepts and their relationships. These abstract ERDs are useful in the early stages of design, such as in planning an electronic resource management system.

Objects :

There are three main objects on an ER Diagram:

1. Entities
2. Relations
3. Attributes

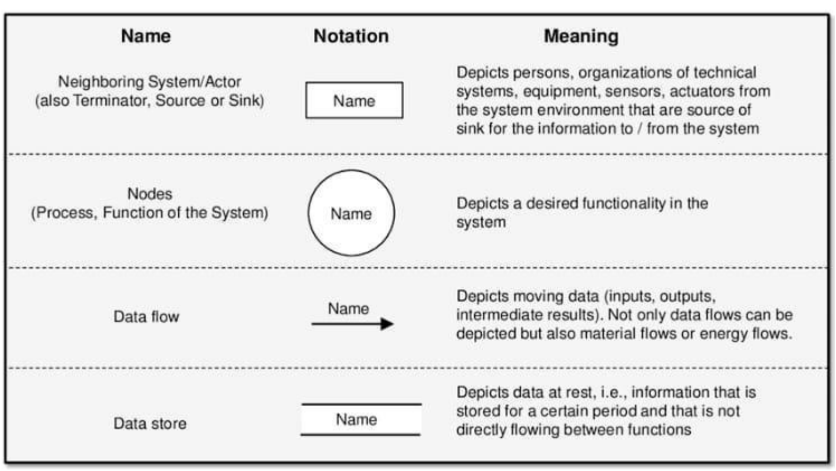
Entities : Entities are objects or concepts within a database, represented as boxes in an ERD. They act as containers for all instances of a specific thing and are equivalent to tables in a relational database.

Attributes : Attributes describe the properties of an entity, such as Supplier Name or Telephone Number. Each attribute holds a single value per entity instance. If multiple values are needed (e.g., multiple phone numbers), this must be accounted for during design.

Relations : Relations define how entities are connected. They show how instances of one entity relate to instances of another, often represented by lines connecting entities in the ERD

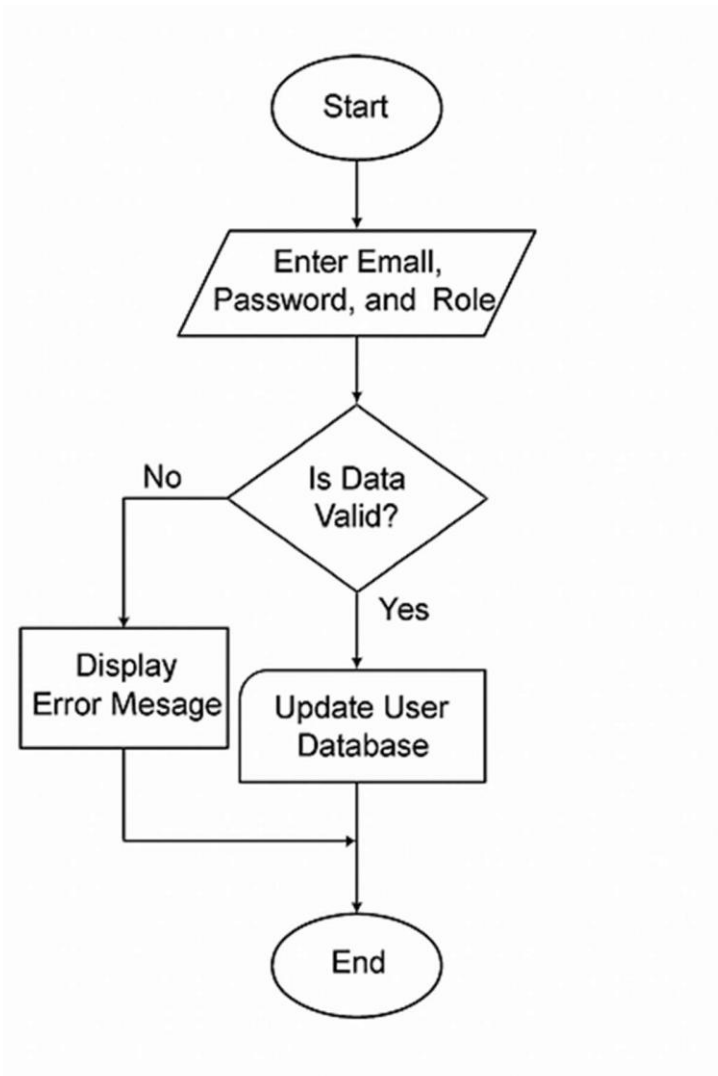
.

3.5.2] System Flow (DFD, Algorithm, flowchart etc..)

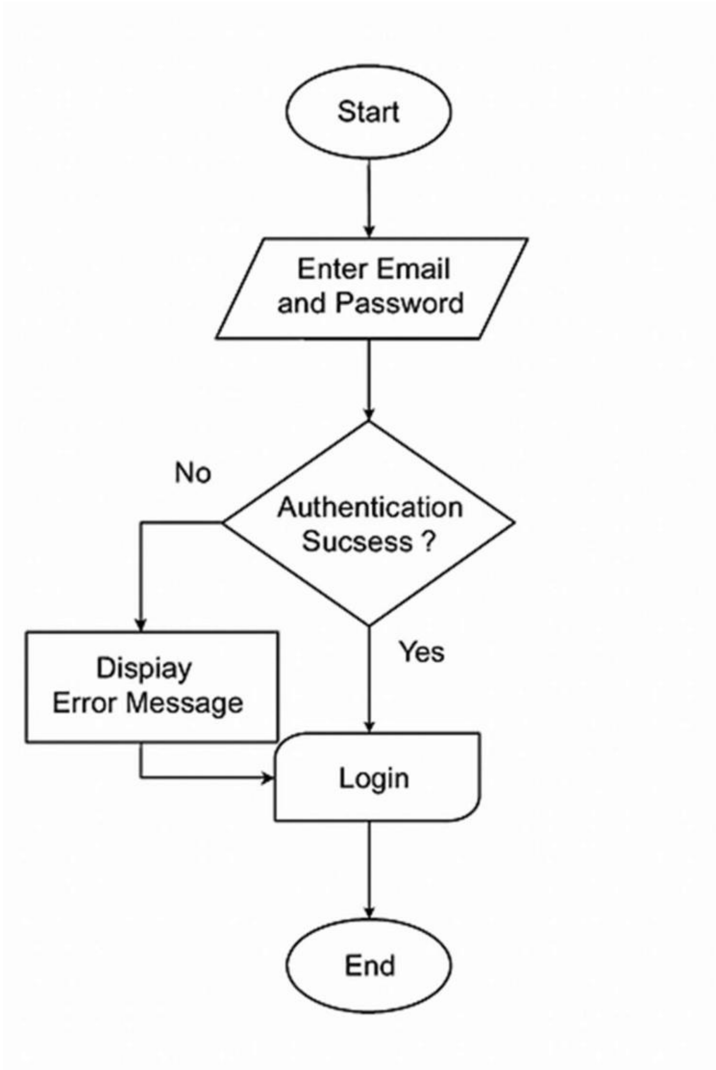


Name	Notation	Meaning
Neighboring System/Actor (also Terminator, Source or Sink)	Name	Depicts persons, organizations of technical systems, equipment, sensors, actuators from the system environment that are source of sink for the information to / from the system
Nodes (Process, Function of the System)	Name	Depicts a desired functionality in the system
Data flow	Name	Depicts moving data (inputs, outputs, intermediate results). Not only data flows can be depicted but also material flows or energy flows.
Data store	Name	Depicts data at rest, i.e., information that is stored for a certain period and that is not directly flowing between functions

For Registration:



For Login:



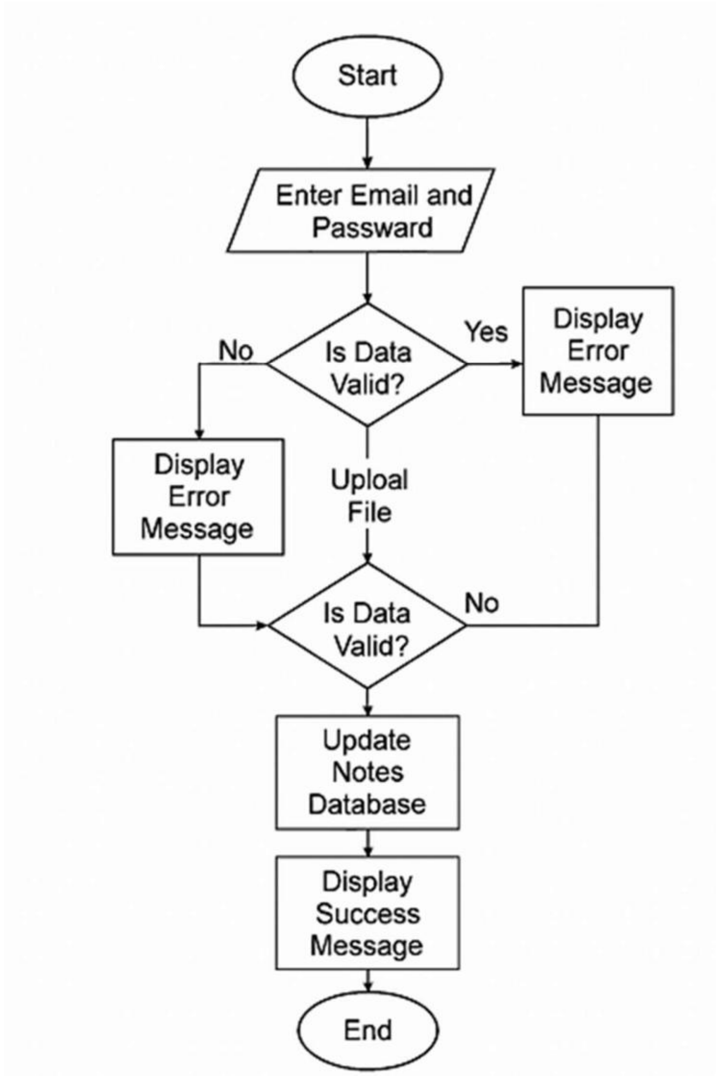
Start
Enter Email
and Password
No
Authentication
Sucsess ?
Yes
Dispiay
Error Message
Login
End

3.5.3 Flow Chart

A flowchart is a graphical representation of a process or system that illustrates the sequence of steps and decisions made within it.

Name	Symbol	Purpose
Terminal	oval	Start /stop/begin/end
Input / output	Parallelogram	Input/output of data
Process	Rectangle	Any processing to be performed can be represented
Desicion box	Diamond	Decision operations that determine which of the alternative paths to be followed
Connector	Circle	Used to connect different parts of flowchart
Flow	Arrows	Joins 2 symbols and also represents flow of execution

Flow chart :


Start
Enter Email and Passward
Is Data Valid?
No
Yes
Display Error Message
Display Error Message
Upload File
Is Data Valid?
No
Update Notes Database
Display Success Message
End

Chapter 4

System Design

4.1 Basic Modules

User Authentication Module:

This module handles secure login and registration for both students and admins. It validates credentials, encrypts passwords, and assigns roles. Upon successful login, users are redirected to their respective dashboards. It ensures only authorized users can access the system and prevents unauthorized access through session management and input validation. This module is critical for maintaining data privacy and system integrity.

Notes Management Module:

This module allows users to upload, categorize, and manage academic notes. Notes are tagged by subject, topic, and file type, making them easy to search and retrieve. Uploaded files are stored securely in the database, and metadata is linked to the uploader's profile. Admins can monitor uploads to ensure quality and relevance. This module forms the core of the platform's academic resource sharing functionality.

Download History Module:

This module tracks every note downloaded by users. It logs the user ID, note ID, and timestamp, storing the data in the Download History table. This ensures accountability and allows admins to monitor usage patterns. It also helps users revisit previously downloaded notes and supports analytics for future enhancements. The module strengthens transparency and system monitoring.

Admin Control Panel:

The admin panel provides tools for managing users, notes, and system activity. Admins can approve or remove notes, and view download statistics. This module ensures the platform remains organized, secure, and free from misuse. It empowers admins to maintain academic integrity and enforce platform policies effectively.

4.2 Data Design

Data designing refers to the process of defining how data will be collected, structured, stored, and managed within a system. It involves creating a blueprint for data architecture that ensures efficient data processing, retrieval, and analysis. Key components include:

1. **Data Models : Designing conceptual, logical, and physical models to represent data and its relationships.**

2. Schema Design : Defining the structure of databases, including tables, fields, and data types.

3. Normalization : Organizing data to reduce redundancy and improve data integrity.

4. Data Governance : Establishing policies for data management, quality, and security.

5. Data Flow : Mapping how data moves through the system, from input to output.

Schema Definition :

In a database management system (DBMS), a schema defines the logical structure of the database, including tables, fields, relationships, and constraints. It serves as a blueprint for organizing and accessing data efficiently. For the

Digital Notes Sharing Website, the schema ensures secure user management, structured note storage, and transparent activity tracking.

1. User Table
   - Name
   - Password
   - Register
   - Role - User

2. Admin Table
   - Name
   - Password
   - Role – Admin

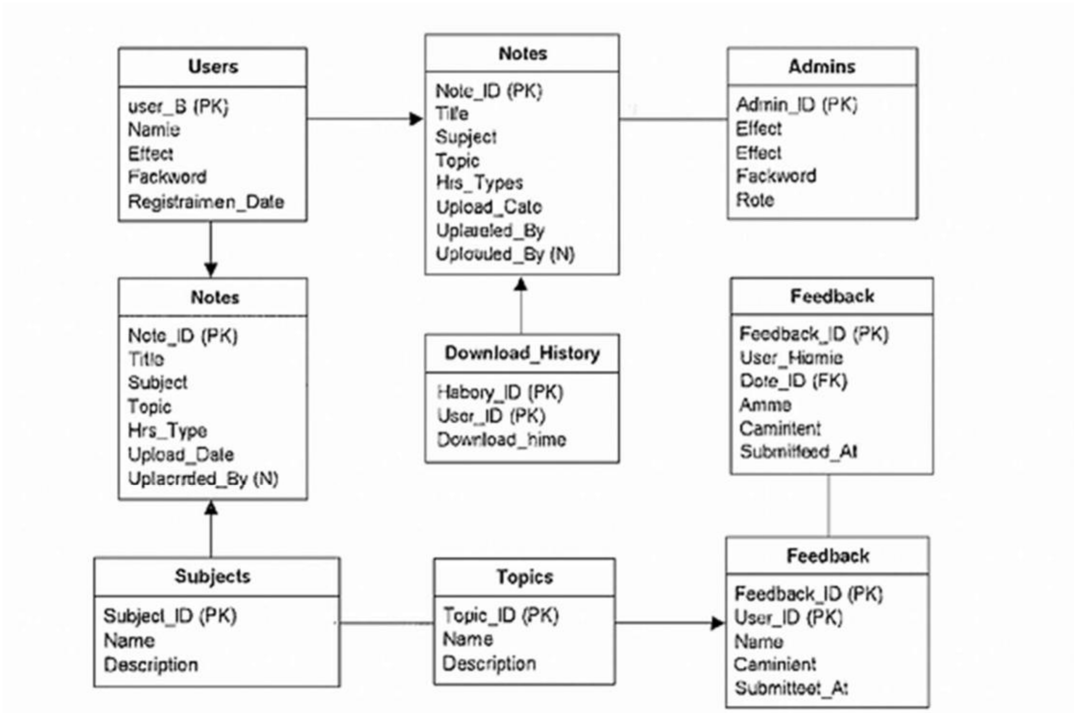
3. Notes table

   - title
   - subject
   - topic
   - file_type
   - file_size note_id
   - upload_date
   - uploaded_by

4. Download_History Table

   - user_id
   - note_id
   - download_date

Final Schema :



4.3 Procedural Design

Procedural design focuses on the logical flow of operations within the system. Each procedure defines a specific task, its input, processing logic, and output. Below are the core procedures implemented in the system:

1. User Registration Procedure: Validates user input (name, email, password, role), checks for duplicate emails, and stores the data in the Users table. Ensures secure onboarding and role-based access.

2. Login Authentication Procedure: Verifies credentials against stored records. If successful, initiates a session and redirects the user to the appropriate dashboard. Handles incorrect login attempts with error messages.

3. Note Upload Procedure: Allows users to upload notes with metadata (title, subject, topic, file type). Validates file format and size, then stores the file and metadata in the Notes table.

4. Download Procedure: Authenticated users can download notes. Each download is logged in the Download_History table with user ID, note ID, and timestamp for tracking and analytics.

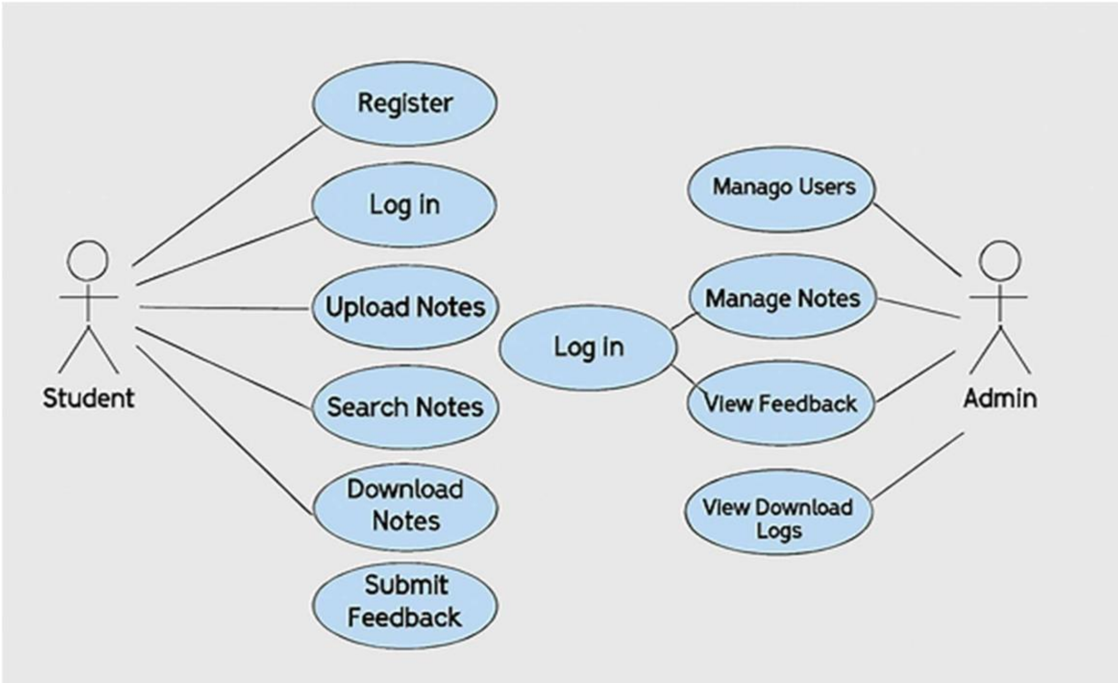
5. Admin Management Procedure: Admins can manage users and notes—approve uploads, delete inappropriate content, and monitor system activity. Ensures platform integrity and academic quality

6. Data Validation Procedure: Ensures all user inputs are sanitized and validated before processing. Prevents SQL injection, maintains data integrity, and enhances security.

7. Error Handling Procedure: Displays appropriate messages for failed actions (e.g., invalid login, upload errors). Logs errors for debugging and improves user experience.

4.4 Use case diagram

A use case diagram visually represents the interactions between users (actors) and a system. It helps identify the functionalities of the system and the relationships between actors and use cases.



4.4.1 Data Structures

Data Structures of Digital Notes sharing website:

Database Tables:

1. Users

• User ID, Name, Email, Password, Role, Registration Date

2. Admins

• Admin ID, Name, Email, Password, Role

3. Notes

• Note ID, Title, Subject, Topic, File Type, File Size, Upload Date, Uploaded By

4. Download_History

- History ID, User ID, Note ID, Download Date

4. Subjects

- Subject ID, Name

Structures:

- Arrays
- Uploaded notes list, Search results
- Linked Lists
- User download history, Feedback entries
- Stacks
- Recently viewed notes, Navigation actions
- Queues
- Upload processing, Admin approval queue
- Trees
- Subject-topic hierarchy, File categorization
- Hash Tables
- User session management, Note search indexing
- Graphs
- User-note interaction mapping, Admin-user relationships Data

Types:

1. Integers

• User ID, Note ID, Feedback ID

2. Varchar

• User name, Email, Note title, Subject name

3. Datetime

• Registration date, Upload date, Download date

4. Decimal

• File size, Rating

5. Boolean / Enum

• User role (Student/Admin), Note approval status

4.4.2 Algorithm Design

Before Login

• Login → Validate email and password → Start session if credentials match → Redirect to user/admin dashboard

• Register → Collect user details → Check for duplicate email → Hash password and store in database → Confirm registration

After Login

- Edit Website Details → Update site title, description, contact info

• Add Subject → Input subject name and description → Store in Subjects table

• Add Topic → Select subject → Input topic name and description → Store in Topics table

• Add Notes → Upload file with metadata → Validate and store in Notes table

• Delete Subject → Select subject → Remove from Subjects table

• Delete Topic → Select topic → Remove from Topics table

• Delete Notes → Select note → Remove from Notes table

• Logout → End admin session → Redirect to homepage

After User Login

• My Downloads → View previously downloaded notes

• Browse Notes → View available notes by subject/topic

Categories → Subjects and topics managed by admin

- Upload -> There Notes

• Logout → End user session → Redirect to homepage

4.5 User interface design

Among all the major components of the system, a major role of the system is played by user interfaces. Interactivity in between system and the user is managed by the interface. User friendliness, integrated colour combination and the well organized components are dependent on it. Without having a user friendly interface, interaction with the system becomes hard. Privileges must be set by the System Administrator to users in different ways. Management of those privileges and presenting them effectively to users is helped by the interfaces.

Security Issues :

Security is crucial for our e-commerce system, covering both the application features and the database. We have a strong user authentication process to prevent unauthorized changes. Maintaining database security is vital, as it supports the entire system. Only authorized users can access and update information by entering the correct password. If a password is wrong, access is denied. The system administrator can change information and manage user permissions, making security a top priority. Application security uses various methods to protect against external threats, ensuring safe transactions and data management.

Chapter 5

Implementation And Testing

5.1 Implementation Approaches

 Requirement Analysis – Project requirements were collected and analyzed to understand user needs and system goals.

 Modular Approach – The project was divided into small modules to make development and testing easier.

 Incremental Development – Features were implemented step by step, starting from basic functionalities to advanced features.

 Database Implementation – A structured database was used to store and manage data efficiently.

 User Interface Design – A simple and user-friendly interface was developed for easy interaction.

 Testing and Deployment – The system was tested for errors and then deployed for final use.

5.2 Coding Details and Code Efficiency

5.2.1/2 Code Efficiency and Test Approach

- Code efficiency means writing optimized code that uses less time and memory while giving correct output.
- Proper coding standards and reusable functions were used to improve performance.
- Regular testing was done during development to reduce errors and improve reliability.

5.2.1 Unit Testing

- Unit testing checks individual modules or functions separately.
- It helps find errors early in development.
- **Example:** Testing the login function to verify correct username and password validation.

5.2.2 Integration Testing

- Integration testing checks whether different modules work together correctly.
- Data flow and communication between modules are tested.
- **Example:** Testing if the login module correctly connects with the database module.

5.2.3Beta Testing

- Beta testing is done by real users before final release.
- Users test the system in real conditions and provide feedback.
- Helps identify usability issues and minor bugs before deployment.

5.3 Modifications and Improvements

• Improved user interface for better usability.

• Fixed bugs identified during testing.

• Optimized database queries for faster performance.

• Enhanced form validation and error messages.

• Improved order confirmation and status display.

Chapter 6

Results and Discussion

6.1 Test Reports

Unit Testing Report : Unit testing was performed to check individual modules separately. Each function was tested with different inputs to ensure correct output. Example modules tested include user login, menu display, order placement, and payment functions. All units worked as expected after fixing minor errors.

Integration Testing Report : Integration testing was carried out to verify that different modules work properly together. The interaction between login, menu, order, and payment modules was tested. Data flow between modules was successful and no major issues were found.

Beta Testing Report : Beta testing was conducted by real users in a real environment before final deployment. Users tested the system and provided feedback regarding usability and performance. Minor improvements were made based on user suggestions.

Overall Result:
The testing process confirmed that the system functions correctly and meets project requirements. The project performed efficiently with successful outputs in all major test scenarios, making it ready for final deployment.

6.2 User Documentation

1.user Guide :

- User login / register
- Create password
- Upload notes
- Uses pre-Uploaded Notes
- Logout

2.Admin Guide

- Admin login
- Manages notes
- Delete Notes
- Approval of User Notes

3.System Requirements

- Web browser (chorme, Brave, Firefox)
- Internet connection
- Any device

Chapter 7:

Conclusions

7.1Conclusion

The project was successfully designed and implemented according to the given requirements. All modules were developed using a structured approach and tested through unit, integration, and beta testing to ensure proper functionality. The system provides a user-friendly interface and performs its tasks efficiently and accurately. Overall, the project achieved its objectives and can be used effectively in real-world scenarios with scope for future improvements and additional features.

7.1.1Significance of the System

The system is significant because it provides an efficient and organized way to perform tasks that were previously done manually. It improves accuracy, saves time, and reduces human errors. The system offers a user-friendly interface, better data management, and faster processing, making the overall operation more reliable and convenient for users.

7.2 Limitations of the System

- Limited features compared to large commercial systems.
- Security features are basic and can be improved in future versions.
- The system may not handle very large amounts of data efficiently.

- User interface customization options are limited.
- Requires regular maintenance and updates for better performance.

Reference

- W3Schools – HTML, CSS, JavaScript, PHP Tutorials
- PHP Official Documentation – https://www.php.net/docs.php
- MySQL Documentation – https://dev.mysql.com/doc/
- MDN Web Docs – JavaScript Guide
- XAMPP Documentation (Apache & MySQL Setup) – https://www.apachefriends.org
- Software Engineering Textbook – Pressman, R.S.

GLOSSARY

• Front End – The user interface of the system that users interact with, designed using HTML, CSS, Bootstrap, and JavaScript.

• Back End – Server-side logic developed using PHP to process data and handle system operations.

• PHP – A server-side scripting language used to create dynamic web pages and manage application logic.

• MySQL – A relational database management system used to store and manage project data.

• Database – An organized collection of data used for storing user and system information.

• JavaScript – A scripting language used to add interactivity and validation in the web application.

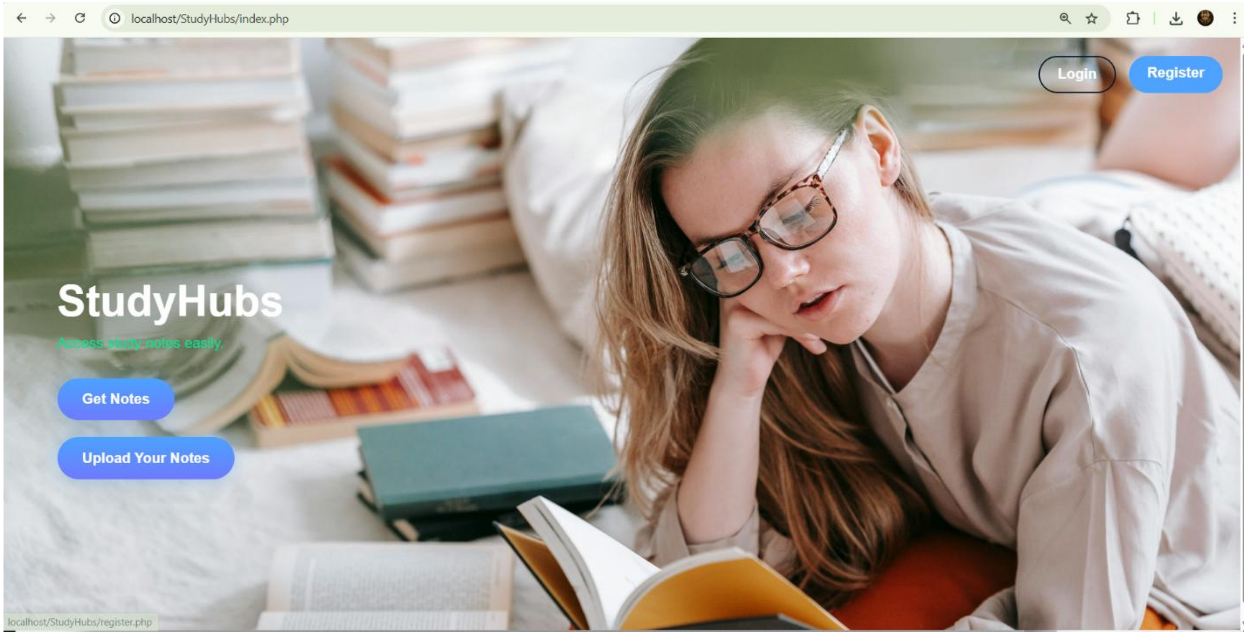
• XAMPP – A local server environment used to run PHP and MySQL during project development and testing.

• User Login – Authentication process that verifies user credentials before allowing system access.

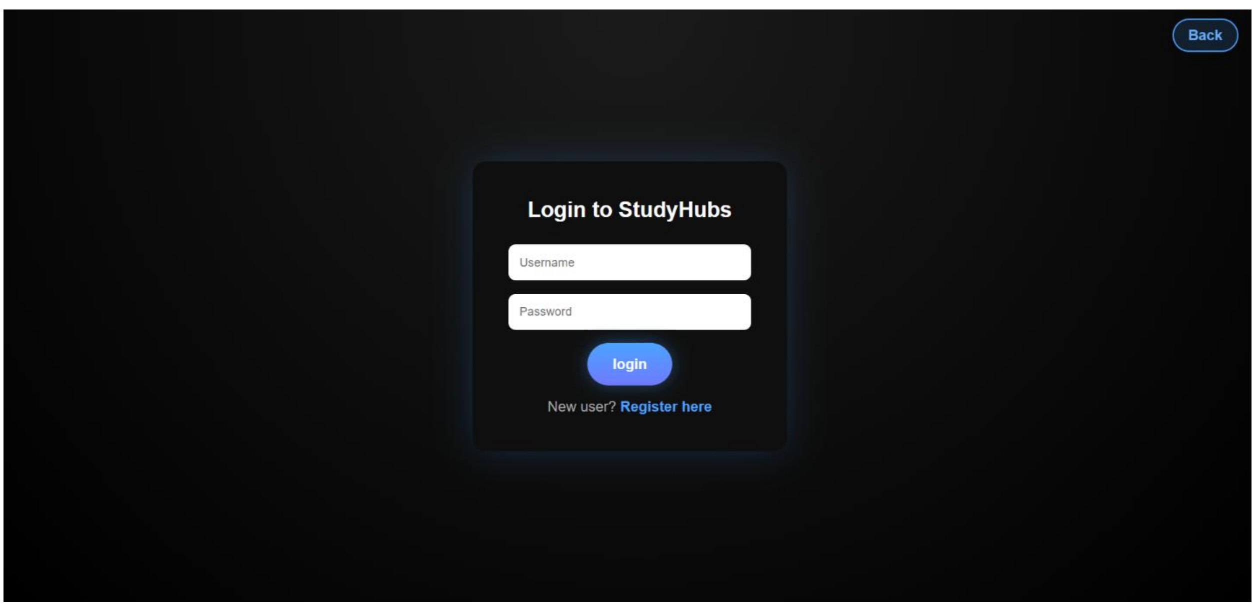
• Testing – Process of checking the system for errors and ensuring all modules work correctly.

APPENDIX A

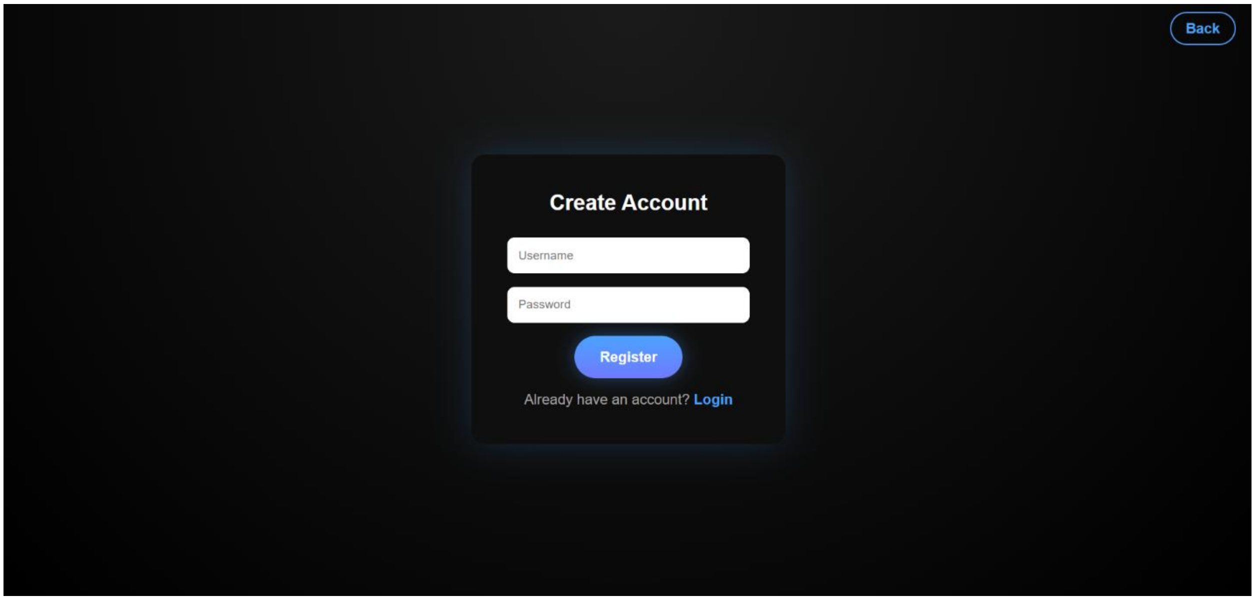
Frontend:



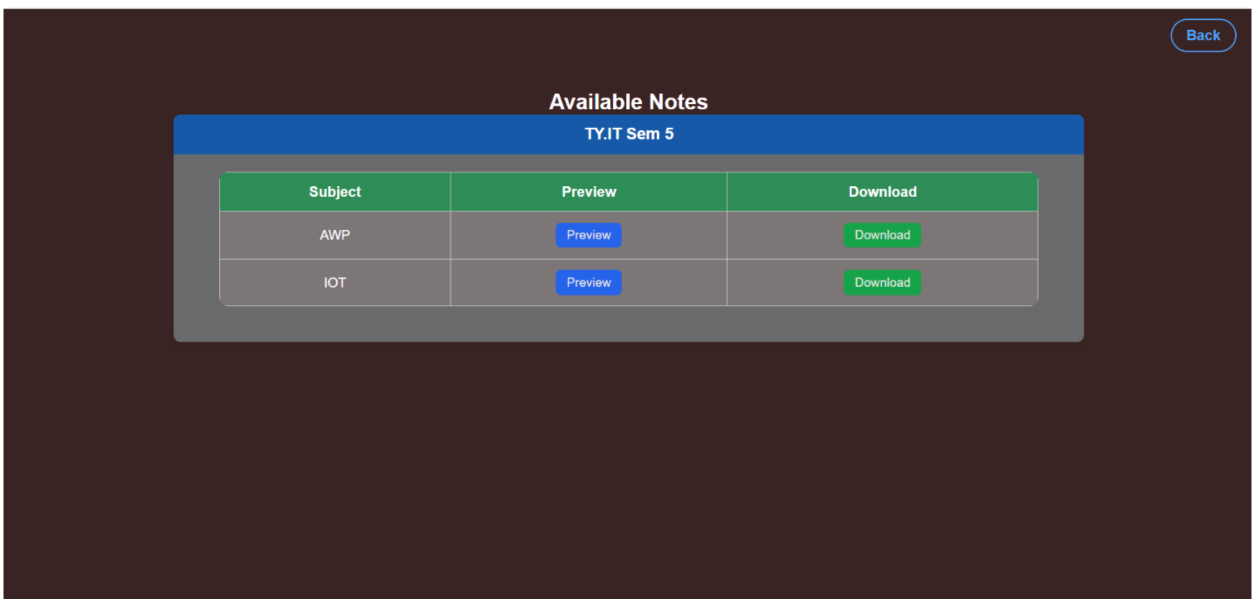
Login:



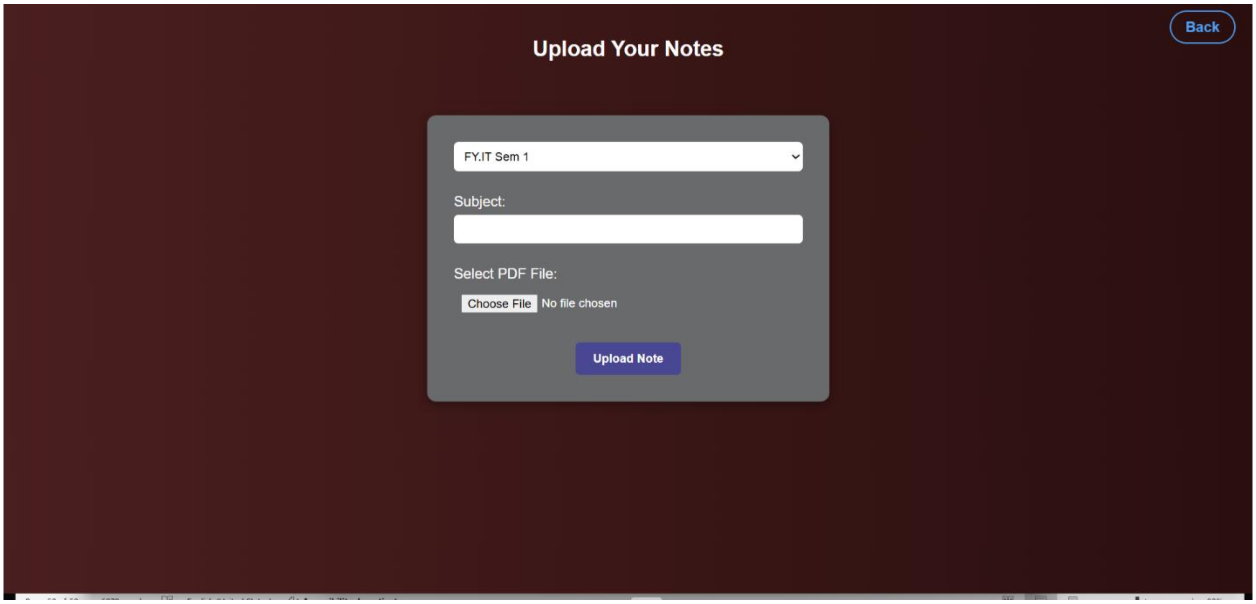
Register page:



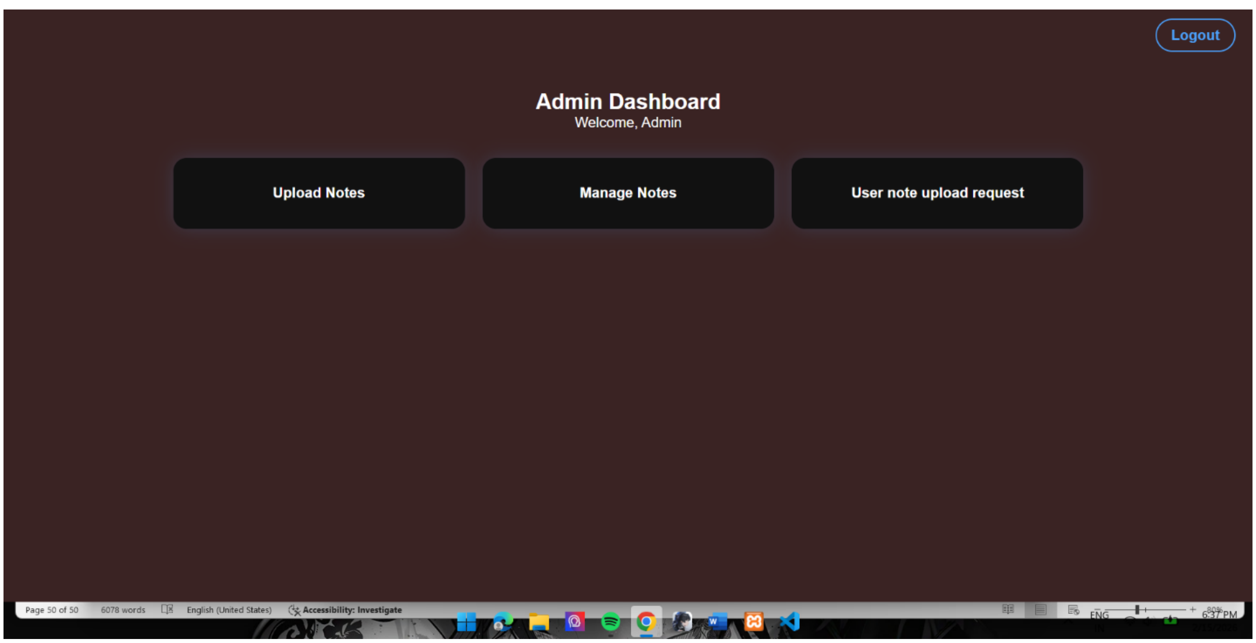
Get Notes(Available Notes):



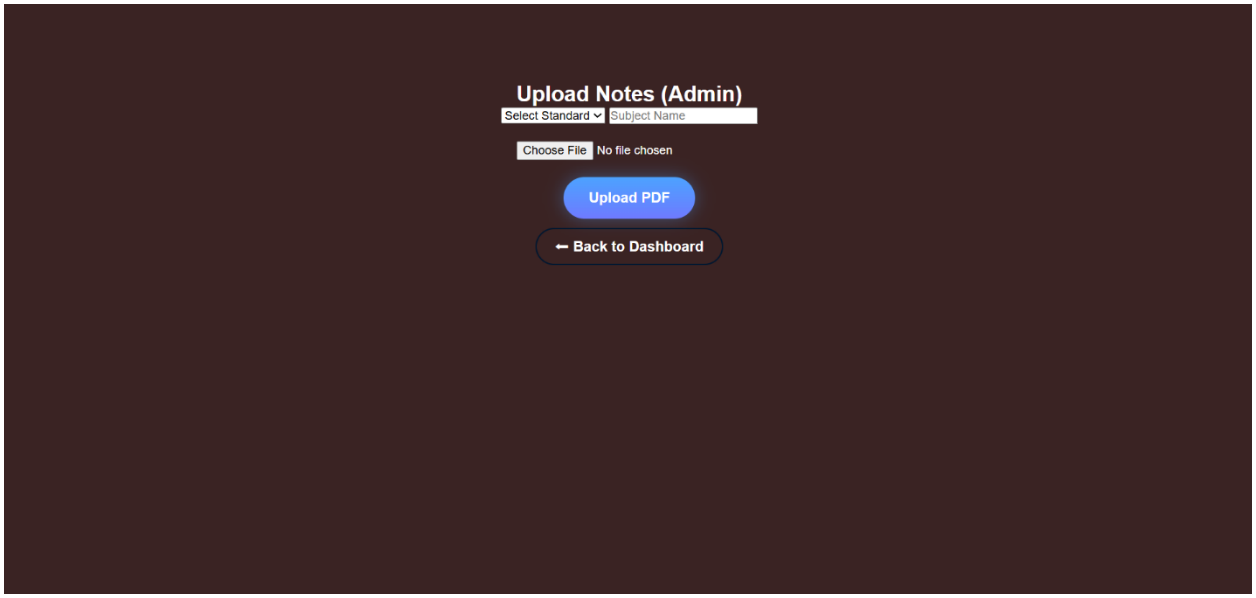
User Note upload Page:



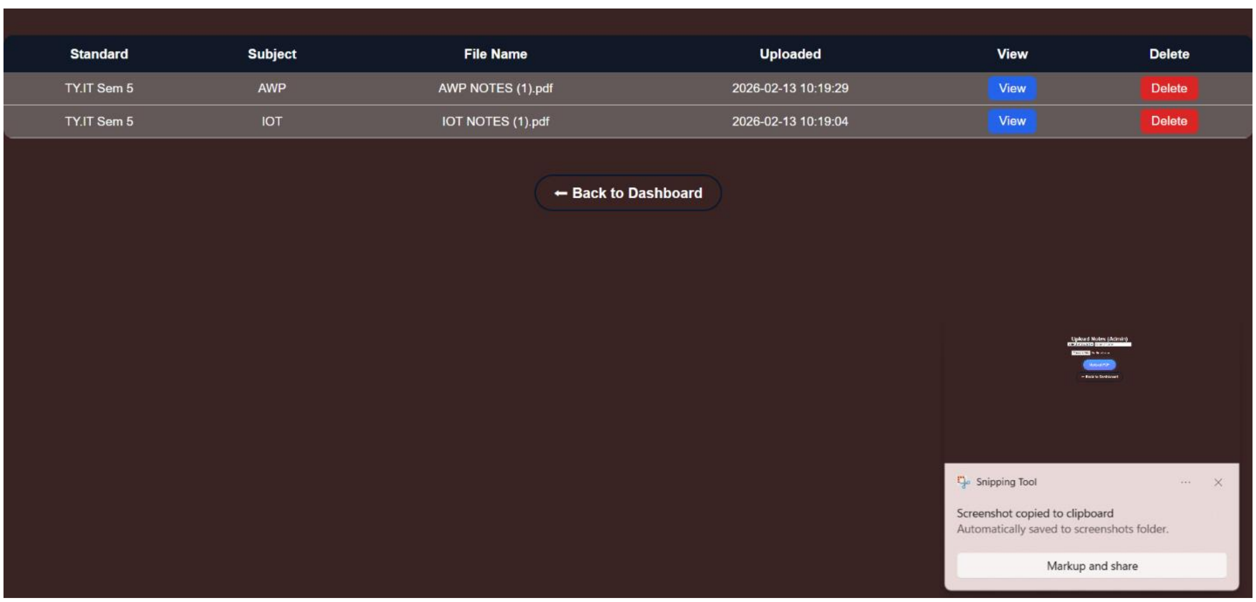
Admin page:



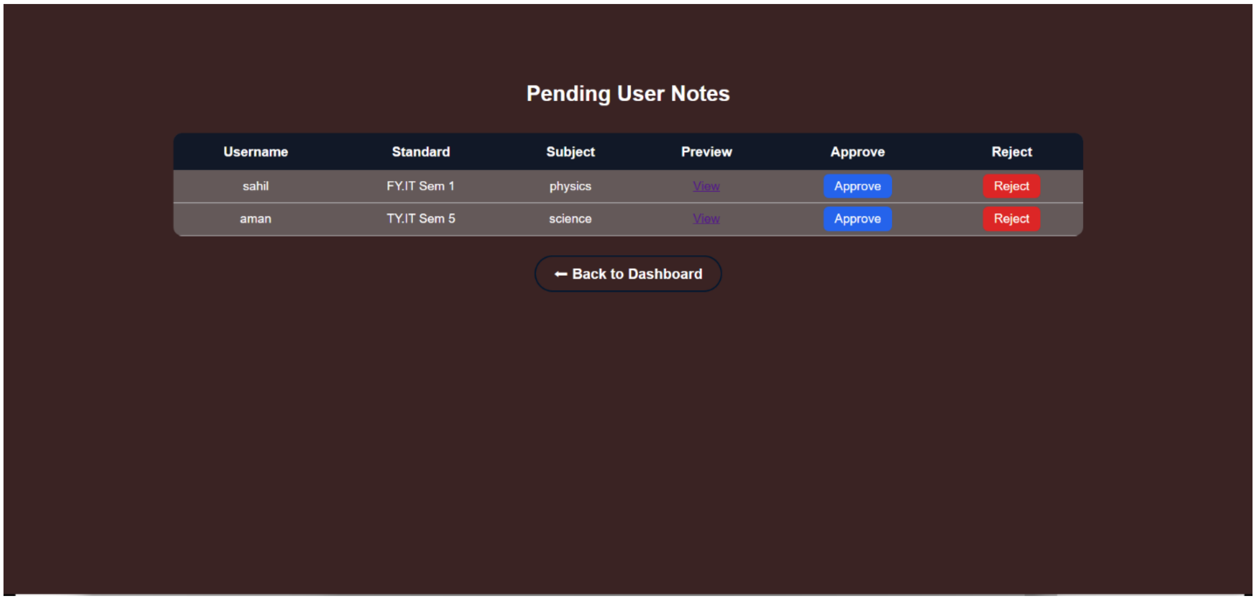
Admin Note Upload page:



Admin manages Notes:



User Notes Approval Page :



APPENDIX B

Frontend code:

```
Index.php: <?php session_start(); ?>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>StudyHubs</title>

<link rel="stylesheet" href="assets/css/style.css">

</head>

<body>

<div class="background-slider"></div>

<div class="navbar">

<?php if(isset($_SESSION['user_id'])): ?>

<a href="logout.php" class="nav-btn login">Logout</a>

<?php else: ?>

<a href="login.php" class="nav-btn login">Login</a>

<a href="register.php" class="nav-btn register">Register</a>

<?php endif; ?>

</div>

<div class="container">
```

```
<div class="left">
<h1>StudyHubs</h1>
<p>Access study notes easily.</p>
<a href="notes.php"><button class="btn main-btn">Get
Notes</button></a>
<a href="upload_note_user.php"><button class="btn main-btn">Upload
Your Notes</button></a>
</div>
</div>
<script>
const images=[
"assets/image/1.jpg","assets/image/2.jpg","assets/image/3.jpg",
"assets/image/4.jpg","assets/image/5.jpg","assets/image/6.jpg"
];
let current=0, slider=document.querySelector(".background-slider");
function changeBackground(){
slider.style.backgroundImage=`url(${images[current]})`;
current=(current+1)%images.length;
}
changeBackground();
setInterval(changeBackground,4000);
</script></body></html>
```

Notes.php: <?php

```
session_start();
include "config.php";
if(!isset($_SESSION['user_id'])){ header("Location: login.php"); exit(); }
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Notes | StudyHubs</title>
<?php echo '<link rel="stylesheet"
href="/StudyHubs/assets/css/style.css?v='.time().'">'; ?>
</head>
<body>
<div class="notes-container">
<h2>Available Notes</h2>
<?php
$result=mysqli_query($conn,"SELECT * FROM notes WHERE
status='approved' ORDER BY standard ASC, uploaded_at DESC");
$current_standard="";
if(mysqli_num_rows($result)>0){
while($row=mysqli_fetch_assoc($result)){
```

```
if($current_standard!=$row['standard']){
if($current_standard!="") echo "</table></div>";
echo "<div class='standard-box'>";
echo "<div class='standard-title'>{$row['standard']}</div>";
echo "<table class='notes-table'>
<tr><th>Subject</th><th>Preview</th><th>Download</th></tr>";
$current_standard=$row['standard'];
}
echo "<tr>
<td>{$row['subject']}</td>
<td><a href='notes/{$row['file_name']}' class='preview-btn' target='_blank'>Preview</a></td>
<td><a href='notes/{$row['file_name']}' class='download-btn' download>Download</a></td>
</tr>";
}
echo "</table></div>";
}else{
echo "<p>No notes uploaded yet.</p>";
}
?>
</div><div class="auth-nav">
```

```
<a href="index.php" class="nav-btn">Back</a>
</div></body></html>
```

<u>Upload User notes.php:</u>

```
<?php
session_start();
include "config.php";
if(!isset($_SESSION['user_id'])){ header("Location: login.php"); exit(); }
if($_SERVER["REQUEST_METHOD"]=="POST"){
$standard=mysqli_real_escape_string($conn,$_POST['standard']);
$subject=mysqli_real_escape_string($conn,$_POST['subject']);
$username=mysqli_real_escape_string($conn,$_SESSION['username'])
;
$file=mysqli_real_escape_string($conn,$_FILES['file']['name']);
$temp_name=$_FILES['file']['tmp_name'];
$folder="notes/".$file;
if(move_uploaded_file($temp_name,$folder)){
$query="INSERT INTO notes
(standard,subject,file_name,username,status)
VALUES ('$standard','$subject','$file','$username','pending')";
mysqli_query($conn,$query);
echo "<script>alert('Note uploaded successfully! Waiting for admin
approval.');</script>";
```

```
}else{
echo "<script>alert('File upload failed!');</script>";
}
}
?>
<!DOCTYPE html>
<html>
<head>
<title>Upload Notes</title>
<link rel="stylesheet" href="assets/css/style.css">
</head>
<body class="sahil">
<h2>Upload Your Notes</h2>
<div class="notes-container1">
<form method="POST" enctype="multipart/form-data">
<select name="standard">
<option value="FY.IT Sem 1">FY.IT Sem 1</option>
<option value="FY.IT Sem 2">FY.IT Sem 2</option>
<option value="SY.IT Sem 3">SY.IT Sem 3</option>
<option value="SY.IT Sem 4">SY.IT Sem 4</option>
<option value="TY.IT Sem 5">TY.IT Sem 5</option>
```

```
<option value="TY.IT Sem 6">TY.IT Sem 6</option>
</select>
<label>Subject:</label>
<input type="text" name="subject" required>
<label>Select PDF File:</label>
<input type="file" name="file" accept="application/pdf" required><br>
<button type="submit" class="nav-btn">Upload Note</button>
</form>
</div>
<div class="auth-nav">
<a href="index.php" class="nav-btn">Back</a>
</div></body></html>
```

Admin Dashboard.php: <?php

```
session_start();
if(!isset($_SESSION['user_id']) || $_SESSION['role']!=='admin'){
header("Location: ../login.php");
exit();
}
?>
<!DOCTYPE html>
```

```
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Admin Dashboard | StudyHubs</title>
<link rel="stylesheet" href="../assets/css/style.css">
</head>
<body>
<div class="notes-container">
<h2>Admin Dashboard</h2>
<p>Welcome, Admin</p>
<div class="notes-grid">
<a href="upload_notes.php" class="note-card">Upload Notes</a>
<a href="manage_notes.php" class="note-card">Manage Notes</a>
<a href="review_user_notes.php" class="note-card">User note upload
request</a>
<div class="auth-nav">
<a href="../logout.php" class="nav-btn">Logout</a>
</div>
</div>
</div>
</body></html>
```

Review user notes.php: <?php

```
session_start();
include "../config.php";
$query="SELECT * FROM notes WHERE status='pending'";
$result=mysqli_query($conn,$query);
?>
<!DOCTYPE html>
<html>
<head>
<title>Review User Notes</title>
<link rel="stylesheet" href="../assets/css/style.css">
</head>
<body>
<div class="notes-container">
<h2>Pending User Notes</h2>
<table class="admin-table">
<tr>
<th>Username</th><th>Standard</th><th>Subject</th>
<th>Preview</th><th>Approve</th><th>Reject</th>
</tr>
<?php while($row=mysqli_fetch_assoc($result)){ ?>
```

```
<tr>
<td><?php echo $row['username']; ?></td>
<td><?php echo $row['standard']; ?></td>
<td><?php echo $row['subject']; ?></td>
<td><a href="../notes/<?php echo $row['file_name']; ?>"
target="_blank">View</a></td>
<td><a href="approve_note.php?id=<?php echo $row['id']; ?>"
class="view-btn">Approve</a></td>
<td><a href="reject_note.php?id=<?php echo $row['id']; ?>"
class="delete-btn">Reject</a></td>
</tr>
<?php } ?>
</table>
<div class="center-btn">
<a href="dashboard.php" class="nav-btn login">⟵ Back to
Dashboard</a>
</div>
</div></body></html>
```

<u>Admin Upload notes.php</u>: <?php

```
session_start();
include "../config.php";
```

```
if(!isset($_SESSION['user_id']) || $_SESSION['role']!=='admin'){

header("Location: ../login.php");

exit();

}

$message="";

if(isset($_POST['upload'])){

$subject=mysqli_real_escape_string($conn,$_POST['subject']);

$standard=mysqli_real_escape_string($conn,$_POST['standard']);

$file=$_FILES['pdf']['name'];

$tmp=$_FILES['pdf']['tmp_name'];

$folder="../notes/";

if(pathinfo($file,PATHINFO_EXTENSION)!=='pdf'){

$message="Only PDF files allowed!";

}else{

if(move_uploaded_file($tmp,$folder.$file)){

mysqli_query($conn,"INSERT INTO notes
(subject,standard,file_name,status)

VALUES ('$subject','$standard','$file','approved')");

$message="PDF uploaded and saved!";

}else{

$message="Upload failed!";
```

```
}}}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Upload Notes | Admin</title>
<link rel="stylesheet" href="../assets/css/style.css">
</head>
<body>
<div class="notes-container">
<h2>Upload Notes (Admin)</h2>
<?php if($message) echo "<p>$message</p>"; ?>
<form method="POST" enctype="multipart/form-data">
<select name="standard" required>
<option value="">Select Standard</option>
<option value="FY.IT Sem 1">FY.IT Sem 1</option>
<option value="FY.IT Sem 2">FY.IT Sem 2</option>
<option value="SY.IT Sem 3">SY.IT Sem 3</option>
<option value="SY.IT Sem 4">SY.IT Sem 4</option>
<option value="TY.IT Sem 5">TY.IT Sem 5</option>
```

```
<option value="TY.IT Sem 6">TY.IT Sem 6</option>
</select>
<input type="text" name="subject" placeholder="Subject Name"
required><br><br>
<input type="file" name="pdf" required><br><br>
<button type="submit" name="upload" class="main-btn">Upload
PDF</button>
</form>
<br>
<a href="dashboard.php" class="nav-btn login">⬅ Back to
Dashboard</a>
</div></body></html>
```

Manage notes.php :

```
<?php
session_start();
include "../config.php";
if(!isset($_SESSION['user_id']) || $_SESSION['role']!=='admin'){
header("Location: ../login.php");
exit();
}?>
<!DOCTYPE html>
<html lang="en">
```

```
<head>

<meta charset="UTF-8">

<title>Manage Notes | StudyHubs</title>

<link rel="stylesheet" href="../assets/css/style.css?v=<?php echo time();
?>">

</head>

<body>

<table class="admin-table">

<tr>

<th>Standard</th><th>Subject</th><th>File Name</th>

<th>Uploaded</th><th>View</th><th>Delete</th>

</tr>

<?php

$result=mysqli_query($conn,"SELECT * FROM notes ORDER BY
uploaded_at DESC");

if(mysqli_num_rows($result)>0){

while($row=mysqli_fetch_assoc($result)){

echo '<tr>

<td>'.$row['standard'].'</td>

<td>'.$row['subject'].'</td>

<td>'.$row['file_name'].'</td>

<td>'.$row['uploaded_at'].'</td>
```

```
<td><a href="../notes/'.$row['file_name'].'" target="_blank" class="view-btn">View</a></td>

<td><a href="delete_notes.php?id='.$row['id'].'" class="delete-btn"

onclick="return confirm(\'Are you sure you want to delete this note?\');">Delete</a></td>

</tr>';

}

}else{

echo "<tr><td colspan='6'>No notes found.</td></tr>";

}

?>

</table>

<br>

<div class="center-btn">

<a href="dashboard.php" class="nav-btn login">⟵ Back to Dashboard</a>

</div>

</body></html>
```

Config.php:

```
<?php

$host = "localhost";

$user = "root";
```

```
$pass = "";
$db   = "studyhubs_db"; // my database name
$conn = mysqli_connect($host, $user, $pass, $db);
if (!$conn) {
    die("Database connection failed");
}
?>
```

<u>Admin/reject user note.php</u> : <?php

```
include "../config.php";
$id = $_GET['id'];
$query = "DELETE FROM notes WHERE id=$id";
mysqli_query($conn, $query);
header("Location: review_user_notes.php");
exit();
?>
```

<u>Login.php</u> : <?php

```
session_start();
include "config.php";
if(isset($_POST['login'])){
$username=mysqli_real_escape_string($conn,$_POST['username']);
```

```
$password=$_POST['password'];

$query=mysqli_query($conn,"SELECT * FROM userdetails WHERE
Username='$username'");

$user=mysqli_fetch_assoc($query);

if($user && password_verify($password,$user['password'])){

$_SESSION['user_id']=$user['id'];

$_SESSION['username']=$user['Username'];

$_SESSION['role']=$user['role'];

if($user['role']==='admin'){

header("Location: admin/dashboard.php");

}else{

header("Location: index.php");

}

exit();

}else{

$error="Invalid login details";

}}

?>

<!DOCTYPE html>

<html lang="en">

<head>
```

```
<meta charset="UTF-8">

<title>Login | StudyHubs</title>

<link rel="stylesheet" href="assets/css/style.css">

</head>

<body class="auth-body">

<div class="auth-box">

<h2>Login to StudyHubs</h2>

<form method="POST">

<?php if(isset($error)) echo "<p style='color:red;'>$error</p>"; ?>

<input type="text" name="username" placeholder="Username"
required>

<input type="password" name="password" placeholder="Password"
required>

<button type="submit" name="login" class="main-btn">Login</button>

</form>

<p class="auth-link">New user? <a href="register.php">Register
here</a></p>

</div><div class="auth-nav">

<a href="index.php" class="nav-btn">Back</a>

</div>

</body>

</html>
```

<u>Logout.php :</u> <?php

```
session_start();

session_unset();

session_destroy();

header("Location: index.php");

exit();

?>
```